

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - III

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Date of Assignment :2-7-2026	Date/Time of Submission :2-7-2026

Stack Application:

Write a program to:

- a. Implement a stack using an array.

Code:

```
import array as arr
stack1=array('i')
si=int(input("Enter size of stack:"))
print(stack1)

def push():
    print("Push Operation")
    if(len(stack1)==si):
        print("Stack Overflow")

    else:
        ele= int(input("Enter Element: "))
        stack1.append(ele)
        print(stack1)

def pop():
    print("Pop Operation")
    if(isEmpty()==0):
        print("stack underflow")
    else:
        stack1.pop()
        print(stack1)

def peek():
```

```

    print("Peek Operation")
    if (isEmpty()==0):
        print("Stack is empty")
    else:
        print(stack1[-1])
def isEmpty():
    if len(stack1)==0:
        return 0
num=0
while(num<5):
    print("1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit")
    num=int (input("select ch: "))
    if (num==1):
        push()
    elif(num==2):
        pop()
    elif(num==3):
        peek()
    elif(num==4):
        isEmpty()
    else:
        print("PROGRAM CLOSED")
print("Himanshu S037")

```

Output:

```

Enter size of stack:2
array('i')
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch: 1
Push Operation
Enter Element: 23
array('i', [23])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch: 2
Pop Operation
array('i')
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch: 24
PROGRAM CLOSED
Himanshu S037

```

Curiosity Questions:

1. What's the difference between `peek()` and `pop()` in stack operations?
2. Why do we call a stack a "LIFO" structure? Can you think of a real-life example of this?
3. Can we store more than one data type (like `int` and `string`) in a stack using array module? Why or why not?
4. What happens if we try to `peek()` into an empty stack?
5. How can we check if the stack is empty before using `pop()`?